# MARK SCHEME · TEACHER COPY

11BR · Topic 4: Processor Fundamentals

## Cambridge AS Level Computer Science 9618

Paper 1  Theory Fundamentals

## Total marks: 75

Official Cambridge mark scheme extracts for each question in the matching exam paper, with question numbers changed to match the exam. Source of each question is shown above its rows.

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**Total: 75 marks**

## Question 1 · May/June 2024 Paper 13, original Q3 (a),(b) [7 marks]

<table>
<tr><th>Question</th><th>Answer</th><th>Marks</th></tr>
<tr><td>1(a)</td><td>1 mark for each correct answer:<br><br>
<table>
<tr><th>Program Number</th><th>Code</th><th>ACC Content</th></tr>
<tr><td>1</td><td>LDM #50<br>INC ACC<br>SUB #1</td><td>50</td></tr>
<tr><td>2</td><td>LDI 51<br>ADD 52</td><td>97</td></tr>
<tr><td>3</td><td>LDR #2<br>LDX 50<br>DEC ACC</td><td>48</td></tr>
<tr><td>4</td><td>LDD 52<br>SUB 54<br>INC ACC</td><td>44</td></tr>
</table></td><td>4</td></tr>
<tr><td>1(b)</td><td>1 mark for each correct answer:<br><br>
<table>
<tr><th>Instruction Number</th><th>Instruction</th><th>ACC Content</th></tr>
<tr><td>1</td><td>LSL #2</td><td>1111 1100</td></tr>
</table></td><td>3</td></tr>
</table>

| | | |
|---|---|---|
| 2 | XOR 100 | **1111 0010** |
| 3 | AND 103 | **0011 0111** |

**Question 2 · May/June 2025 Paper 12, original Q1 (a),(b),(c) [9 marks]**

| Question | Answer | Marks |
|---|---|---|
| 2(a) | **1 mark** for each correct row | **3** |

| Incorrect statement number | Corrected statement |
|---|---|
| **1** | The Program Counter (PC) stores the **address of** the next instruction to be fetched from memory. |
| **3** | The Control Unit (CU) sends signals to other components on the **control** bus. |
| **4** | The Memory Data Register (MDR) **holds data to be stored in** the memory address in the MAR. // The Memory Data Register (MDR) **holds data read from** the memory address in the MAR |

| | | |
|---|---|---|
| 2(b) | **1 mark** for name, **1 mark** for corresponding role<br><br>e.g.<br>• Current Instruction Register // CIR<br>• To store the instruction to be decoded / executed next<br><br>• Status register // SR<br>• To contain bits that can be referenced individually to indicate a state or event<br><br>• Interrupt register<br>• To store details of any interrupts that have occurred | **2** |
| 2(c) | **1 mark** each to **max 4**<br><br>• At the start / end of FE cycle the interrupt register is checked<br>• The priority of any interrupts waiting is checked<br>• if the priority of the interrupt is higher than the current process<br>• the contents of the registers are stored on the stack<br>• The relevant Interrupt Service Routine (ISR) / interrupt handler is called to process the interrupt<br>• When the ISR has finished, a further check is made for higher priority interrupts<br>• if no more interrupts of higher priority, the register contents are restored and the next FE cycle continues | **4** |

**Question 3 · May/June 2025 Paper 12, original Q7 (a),(b) [8 marks]**

| Question | Answer | Marks |
|---|---|---|

**Total: 75 marks**

| | | |
|---|---|---|
| 3(a)(i) | **1 mark** for each shaded section | **3** |

| Instruction address | ACC | Memory address | | | | |
|---|---|---|---|---|---|---|
| | | 10 | 11 | 12 | 13 | 14 |
| | | 12 | 11 | 10 | 22 | 22 |
| 100 | 12 | | | | | |
| 101 | 22 | | | | | |
| 102 | | | 22 | | | |
| 103 | | | | | | |
| 104 | | | | | | |
| 107 | | 22 | | | | |
| | | | | | | |

| | | |
|---|---|---|
| 3(a)(ii) | **1 mark** from:<br><br>• The number 10 will be loaded into the ACC instead of 12 / contents of address 10<br>• The addition will give 20 not 22 so the comparison will fail and result in an infinite loop | **1** |

| | | |
|---|---|---|
| 3(a)(iii) | **1 mark** for name of mode, **1 mark** for matching description<br><br>e.g.<br>• Indirect<br>• The operand points to the memory location that contains the address of the data<br><br>• Indexed<br>• The address of the data is formed by adding the contents of the Index Register (IX) to the operand<br><br>• Relative<br>• The address is calculated using its distance from a base address | **2** |
| 3(b)(i) | **1 mark** for:<br><br>1111 1111 | **1** |
| 3(b)(ii) | **1 mark** for:<br><br>0000 0011 | |

**Question 4 · May/June 2025 Paper 13, original Q5 (a),(b) [9 marks]**

| Question | Answer | Marks |
|---|---|---|
| 4(a)(i) | **1 mark** for each shaded section | **3** |

**Total: 75 marks**

| Instruction address | ACC | Memory address | | | |
|---|---|---|---|---|---|
| | | 50 | 51 | 52 | 53 |
| | | 47 | 48 | 49 | 50 |
| 500 | 50 | | | | |
| 501 | 49 | | | | |
| 502 | | | | | |
| 504 | | | | | |
| 501 | 48 | | | | |
| 502 | | | | | |
| 503 | | | | | |
| 505 | | 48 | | | |
| 506 | 38 | | | | |
| 507 | | | 38 | | |
| | | | | | |
| | | | | | |

| | | |
|---|---|---|
| 4(a)(ii) | **1 mark** for addressing mode, **1 mark** for matching description<br><br>• Direct<br>• The operand is the address of the data<br><br>• Indirect<br>• The operand points to the memory location which is the address of the data<br><br>• Indexed<br>• The address of the data is formed by adding the contents of the Index Register (IX) to the operand | **4** |
| 4(b) | **1 mark** for each correct instruction<br><br>e.g.<br>AND B00000000 / #0 / &0<br>OR B10000000 / #128 / &80 | **2** |

**Question 5 · Oct/Nov 2023 Paper 12, original Q9 (a),(b) [8 marks]**

| Question | Answer | Marks |
|---|---|---|
| 5(a) | **1 mark** for each bullet point<br><br>• `IN` - Input **and** output of data<br>• `ADD` - Arithmetic operations<br>• `JPE` - Unconditional and conditional instructions<br>• `CMI` - Compare instructions | **4** |

| | | |
|---|---|---|
| 5(b) | **1 mark** for each set of shaded rows | **4** |

| Instruction address | ACC | IX | Memory address | | | | Output |
|---|---|---|---|---|---|---|---|
| | | | 100 | 101 | 110 | 111 | |
| | | | 0 | 0 | 51 | 65 | |
| 10 | | 0 | | | | | |
| 11 | 49 | | | | | | |
| 12 | | | | 49 | | | |
| 13 | 51 | | | | | | |
| 14 | | | | | | | |
| 15 | | | | | | | |
| 16 | 49 | | | | | | |
| 17 | 65 | | | | | | |
| 18 | | 1 | | | | | |
| 19 | | | 65 | | | | |
| 20 | | | | | | | |
| 13 | 65 | | | | | | |
| 14 | | | | | | | |
| 15 | | | | | | | |
| 21 | | | | | | | A |
| 22 | | | | | | | |

**Total: 75 marks**

## Question 6 · Oct/Nov 2023 Paper 13, original Q9 (a),(b),(c),(d) [13 marks]

<table>
<tr><th>Question</th><th>Answer</th><th>Marks</th></tr>
<tr><td>6(a)(i)</td><td>1 mark for each bullet point (max 3).<br><br>• Control unit (CU)<br>• Arithmetic and Logic Unit (ALU)<br>• Immediate Access Store (IAS)<br>• (System clock)</td><td>3</td></tr>
<tr><td>6(a)(ii)</td><td>1 mark for each bullet point (max 2).<br><br>• Special purpose registers hold the status of a program whereas general purpose registers hold the temporary data while performing operations.<br>• Special purpose registers are specialised for a specific use, whereas general purpose registers are used for any purpose.<br>• General purpose registers can be used by most instructions, whereas special purpose can only be used by certain instruction</td><td>2</td></tr>
<tr><td>6(b)</td><td>1 mark for each correct answer (shaded cells)<br><br><table><tr><th>Step</th><th>Description</th></tr><tr><td>PC ← [PC] + 1</td><td>The address in PC is incremented.</td></tr><tr><td>MDR ← [[MAR]]</td><td>The data in the location pointed to by the MAR is copied to the MDR.</td></tr></table></td><td>4</td></tr>
</table>

| | | | |
|---|---|---|---|
| | | MAR ← [PC] | The **contents** of PC are copied to the MAR. |
| | | CIR ← [MDR] | The contents of MDR are copied into CIR. |
| 6(c) | **1 mark** for an appropriate example.<br><br>Examples:<br>• Division by zero<br>• Runtime error<br>• Out of memory bounds<br>• Program requesting an external device / input<br>• Buffer overflow | | **1** |
| 6(d) | **1 mark** for each letter in the correct position (2, 4 and 6)<br><br>**D**<br>**A**<br>**B** | | **3** |

**Question 7 · Oct/Nov 2025 Paper 12, original Q3 (a),(b),(c) [10 marks]**

| Question | Answer | Marks |
|---|---|---|
| 7(a)(i) | **1 mark**<br><br>• The value of the operand is an offset value which is added to another base value to give the address from which the contents are loaded to the accumulator | **1** |

| | | |
|---|---|---|
| 7(a)(ii) | **1 mark** per bullet point, **max 2** marks<br><br>• Program counter (PC)<br>• Memory Data Register (MDR)<br>• Memory Address Register (MAR)<br>• Current Instruction Register (CIR)<br>• Status Register | **2** |
| 7(b) | **1 mark** for each correct value, **max 3** marks | **3** |
| 7(c) | **1 mark** per bullet point, **max 2** marks for each<br><br>Clock Speed:<br>• Processor can perform more F-E cycles per second<br>• ... so more instructions / data can be processed each second<br><br>Cache Memory:<br>• Can store more of the most frequently used instructions<br>• … which reduces the need to access slower RAM | **4** |

| Instruction | Value in ACC |
|---|---|
| LDM #98 | 98 |
| LDI 101 | 8 |
| LDX 100 | 32 |

**Total: 75 marks**

## Question 8 · Oct/Nov 2025 Paper 13, original Q6 (a),(b),(c) [11 marks]

<table>
<tr><th>Question</th><th>Answer</th><th>Marks</th></tr>
<tr><td>8(a)</td><td>1 mark for each correct row<br><br>
<table>
<tr><th>Current contents of the ACC</th><th>Instruction</th><th>Contents of the ACC after the execution of the instruction</th></tr>
<tr><td>0000 1111</td><td>AND 101</td><td>0000 1110</td></tr>
<tr><td>0000 0000</td><td>LDM #100</td><td>0110 0100</td></tr>
<tr><td>0000 0001</td><td>XOR &F1</td><td>1111 0000</td></tr>
<tr><td>0001 0001</td><td>CMP 101</td><td>0001 0001</td></tr>
</table></td><td>4</td></tr>
<tr><td>8(b)</td><td>1 mark for each bullet point, max 4 marks<br><br>
• MAR stores the address of the next instruction / data to be read from / or written to memory<br>
• The address is received from the Program Counter (PC)<br><br>
• The MDR stores the data / instruction in the address stored in the MAR which has been read / written<br>
• The instruction passes to the CIR for decoding and executing</td><td>4</td></tr>
</table>

| | | |
|---|---|---|
| 8(c) | **1 mark** for each bullet point, **max 3** marks<br><br>• Data movement<br>• Arithmetic operations<br>• Conditional and unconditional (jump) instructions | **3** |

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